

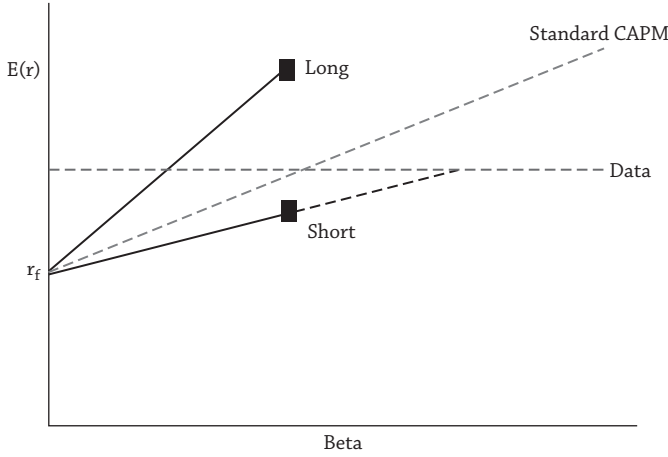


Figure 10.12

“Data” line and levers it up to the square marked “Long.” The short position in the high-beta portfolio is marked “Short.” The *BAB* portfolio does not take the entire position in the high-beta portfolio; it combines it with the risk-free asset to obtain the position marked Short. In effect, the Long and Short portfolios are unit beta positions in the low- and high-beta portfolios.

Frazzini and Pedersen use just two beta portfolios in creating their *BAB* factor. They have little choice. In Figure 10.10, the first quintile portfolio has a pre-ranking beta close to zero. Levering up this portfolio results in a position close to infinity. Thus, one is forced to create very small numbers of portfolios—two or three at most—in the *BAB* factor. One advantage of the volatility portfolios is that they can be directly traded without using the risk-free asset because there are pronounced differences in expected returns, not only volatilities, across the volatility quintiles.

Volatility Factor

I construct a volatility factor, *VOL*, similar to Frazzini and Pedersen’s *BAB*:

$$VOL_{t+1} = \sigma_{\text{target}} \times \left(\frac{r_{L,t+1} - r_f}{\sigma_{L,t}} - \frac{r_{H,t+1} - r_f}{\sigma_{H,t}} \right), \quad (10.18)$$

where $\sigma_{L,t}$ and $\sigma_{H,t}$ are the pre-ranking volatilities of the low- and high-volatility portfolios. While the *BAB* factor scales to unit betas, the *VOL* factor scales to a target volatility. I use the first and fifth quintile portfolios with returns $r_{L,t}$ and $r_{H,t}$, respectively. I set the target volatility $\sigma_{\text{target}} = 15\%$.